

May 20.

Q1: What estimates are good enough to get convergence?

Q2: What is convergence?

A Ricci flow is a 1-parameter family of metrics.

→ Convergence of a sequence of metrics.

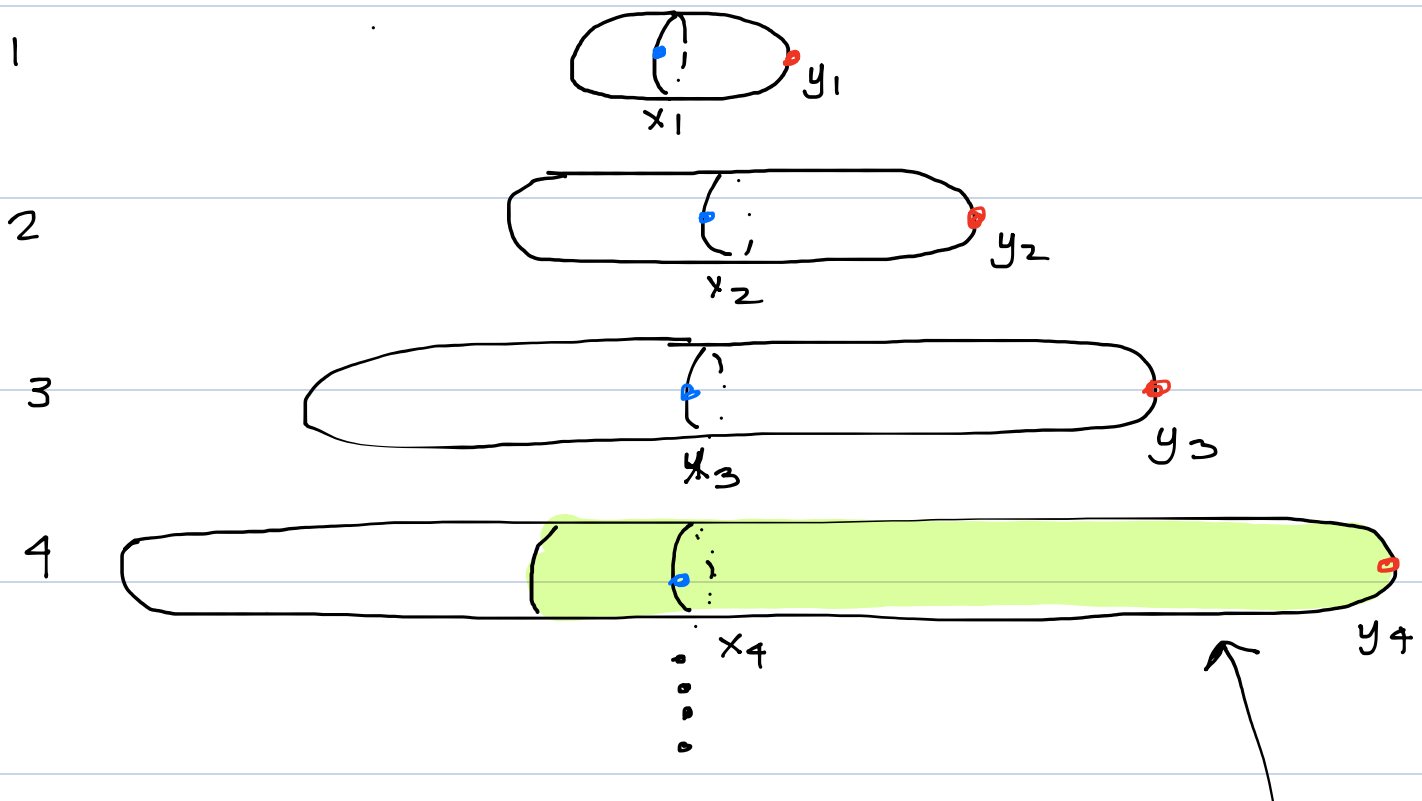
Choose $t_i \rightarrow T$.

Definition (Smooth Cheeger-Gromov convergence)
in the pointed sense.

$$\left(M_i^n, g_i, x_i \right)_{i=1}^{\infty}$$

↪ Riem. metric on M_i , $x_i \in M_i$ base point.

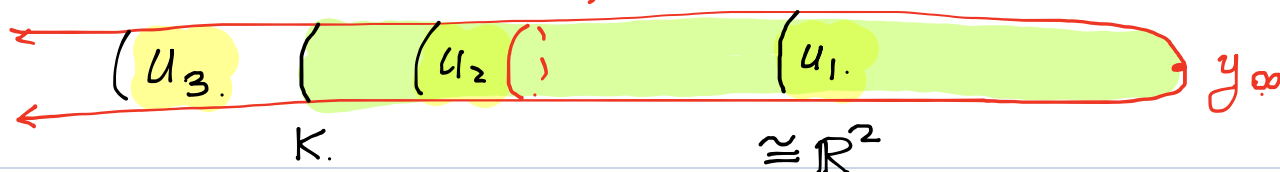
Example $i = 1, 2, 3, 4, \dots$



Limit $(M_\infty^n, g_\infty, x_\infty)$

Another limit $(M_\infty^n, g_\infty, y_\infty)$

$$\cong S^1 \times \mathbb{R}$$



Def. (Pt'ed CG C^∞ conv.)

Let $\{(M_i^n, g_i(t), x_i)\}_{i=1}^\infty$ be a sequence of RFs on common time interval $t \in [\alpha, \omega]$.

This converges to $(M_\infty^n, g_\infty(t), x_\infty)$

$\hookrightarrow$ complete RF

if (def):

There $\exists$ exhaustion $\{\mathcal{U}_i\}_{i=1}^\infty$ of M_∞ by open sets with $x_\infty \in \mathcal{U}_i$ and $\overline{\mathcal{U}_i} \subset \mathcal{U}_{i+1}$.

$\exists$ diffeomorphism Φ_i .

$$\Phi_i : \mathcal{U}_i \xrightarrow{\subset M_\infty} V_i := \Phi_i(\mathcal{U}_i) \subset M_i$$

such that $\Phi_i(x_\infty) = x_i$.

$\forall$ compact $K \subset M_\infty$ and $\forall$ integer $k \geq 0$.

$$\Phi_i^* \underset{\substack{\uparrow \\ \text{on } M_i}}{g_i(t)} \rightarrow g_\infty(t)$$

$\underbrace{\hspace{10em}}$
on M_∞
in $C^k(K)$ uniformly for $t \in [\alpha, \omega]$.

Reference: Hamilton 1995 American J. of Math.
Compactness.

exposition: M.S.M. series of AMS. vol 135.

CGH compactness theorem.

Let $(M_i^n, g_i(t))$, $t \in [\alpha, \omega]$, be a seq. of RFs. complete.

such that:

(1) $\exists C$: $\sup_{M_i^n \times [\alpha, \omega]} |Rm(g_i(t))| \leq C$.

(2) $\exists t_0 \in [\alpha, \omega]$, $x_i \in M_i^n$ & $\kappa > 0$.

such that

$$\text{Vol } B_1(x_i, t_0) \geq \kappa.$$

Then $\exists$ subseq s.t. the pt'd RFs.

$$(M_i^n, g_i(t), x_i), t \in [\alpha, \omega],$$

converges in the C^∞ Cheeger-Gromov-Hamilton sense to a complete RF.

$$(M_\infty^\infty, g_\infty(t), x_\infty), t \in [\alpha, \omega],$$

$$\sup_{M_\infty^\infty \times [\alpha, \omega]} |Rm(g_\infty(t))| \leq C.$$

How can we apply this?

Theorem. (Perelman). [Proof discussed next time].

Let $(M^n, g(t)), t \in [0, T), 0 < T < \infty$.

For any $\rho > 0$, there exists $R = R(g_0, T, \rho) > 0$.

such that:

Suppose.

$R(x, t_0) \leq r^{-2}$ in $B_r(x_0, t_0)$, where $0 < r \leq \rho$.

Then. $\text{Vol}_{g(t_0)} B_r(x_0, t_0) \geq R r^n$.

"No local collapsing".

Perelman's NLC takes care of cond. (2).
in the cptness thm.

What about condition (1)?

Let $(M^3, g(t)), t \in [0, T)$, with $\text{Ric}(g(0)) > 0$,
be a RF def on a maximal time interval.

We know $T < \infty$,

- we are not normalizing the RF,

- $\sup R = \infty$.

$M^3 \times [0, T)$

Let $T_i \nearrow T$.

$\sup_{\mathcal{M}^3 \times [0, T_i]} R. \rightarrow \infty$

Let $(x_i, t_i) \in \mathcal{M}^3 \times [0, T_i]$ assume the max of R ,

So

$$R_i = R(x_i, t_i) = \sup_{\mathcal{M}^3 \times [0, t_i]} R. = \sup_{\mathcal{M}^3 \times [0, T_i]} R. \rightarrow \infty$$

Use R_i to rescale the RF $g(t)$.

Define

$$g_i(t) = R_i g(t_i + R_i^{-1} t)$$

is a seq of rescaled RFs.

$g_i(t)$ is defined for $[-R_i t_i, 0]$

$\forall \alpha < 0, [\alpha, 0] \subset [-R_i t_i, 0]$.

for i large enough.

$\nearrow -\infty$
could go to
 $R_i(T - t_i)$

Simply apply CGH compactness thm.

$g_i(t)$ has $R_{g_i(t)} \leq 1$ and $R(x_i, 0) = 1$.

so $|Rm_{g_i(t)}| \leq C$, so (1) holds.

By Perelman's NLC,

since $R(x, t_i) \leq R_i = r^{-2}$

in $B_r(x_i, t_i) \subset M$.

(where $r = R_i^{-1/2} \rightarrow 0$),

we have $\text{Vol}_{g(t_i)} B_r(x_i, t_i) \geq \kappa r^3$.

By rescaling, this says

$$\boxed{\text{Vol}_{g_i(0)} B_1(x_i, 0) \geq \kappa}, \text{ so cond (2) holds.}$$

Recall $g_i(t) = R_i g(t_i + R_i^{-1} t)$

So $(M^3, g_i(t), x_i) \rightarrow (M_\infty^3, g_\infty(t), x_\infty)$
 $t \in (-\infty, 0]$.

complete RF with $R_{g_\infty(t)} \leq 1$.

and $R_{g_\infty(0)}(x_\infty) = 1 > 0$.

Claim: $(M_\infty^3, g_\infty(t))$ is a shrinking SSF.

Since $\partial_t R = \Delta R + 2|\text{Ric}|^2$, by the strong maximum principle to conclude

$$\boxed{R_{g_\infty(t)}(x) > 0},$$

Recall the key pinching improves estimate:

$$\frac{|Ric - \frac{1}{3}Rg|^2}{R^{2-\varepsilon}} \leq C(\varepsilon) \text{ on } M^3 \times [0, T).$$

from $Ric \geq \delta Rg$ (preserved), $\varepsilon = 2\delta^2$,

$g(t)$ satisfies.

$$\frac{|Ric - \frac{1}{3}Rg|^2}{R^2} \leq C(\varepsilon) R^{-\varepsilon}$$

So $g_i(t)$ satisfies

$$\frac{|Ric(g_i) - \frac{1}{3}R_{g_i}g_i|^2}{R_{g_i}^2} \leq C(\varepsilon) \underbrace{R_i^{-\varepsilon}}_{\substack{\text{ } \\ \text{ } }} R_{g_i}^{-\varepsilon}$$

We can take a limit: get from CGH conv:

$$\frac{|Ric(g_\infty) - \frac{1}{3}R_{g_\infty}g_\infty|^2}{R_{g_\infty}^2} \leq 0 \cdot R_{g_\infty}^{-\varepsilon} = 0.$$

So (M_∞^3, g_∞) is Einstein, so it's a SSF.

So M_∞^3 is compact.

Since M_∞^3 is compact, M_∞^3 is diffeomorphic to M^3 . So M^3 admits a const positive sectional curvature metric.

To do: Perelman's NLC.

Swept under the rug:

W.-X. Shi's local derivative of
curvature estimates.

- coarse estimate.
- Bernstein estimate, done globally by Bando.
for RF.